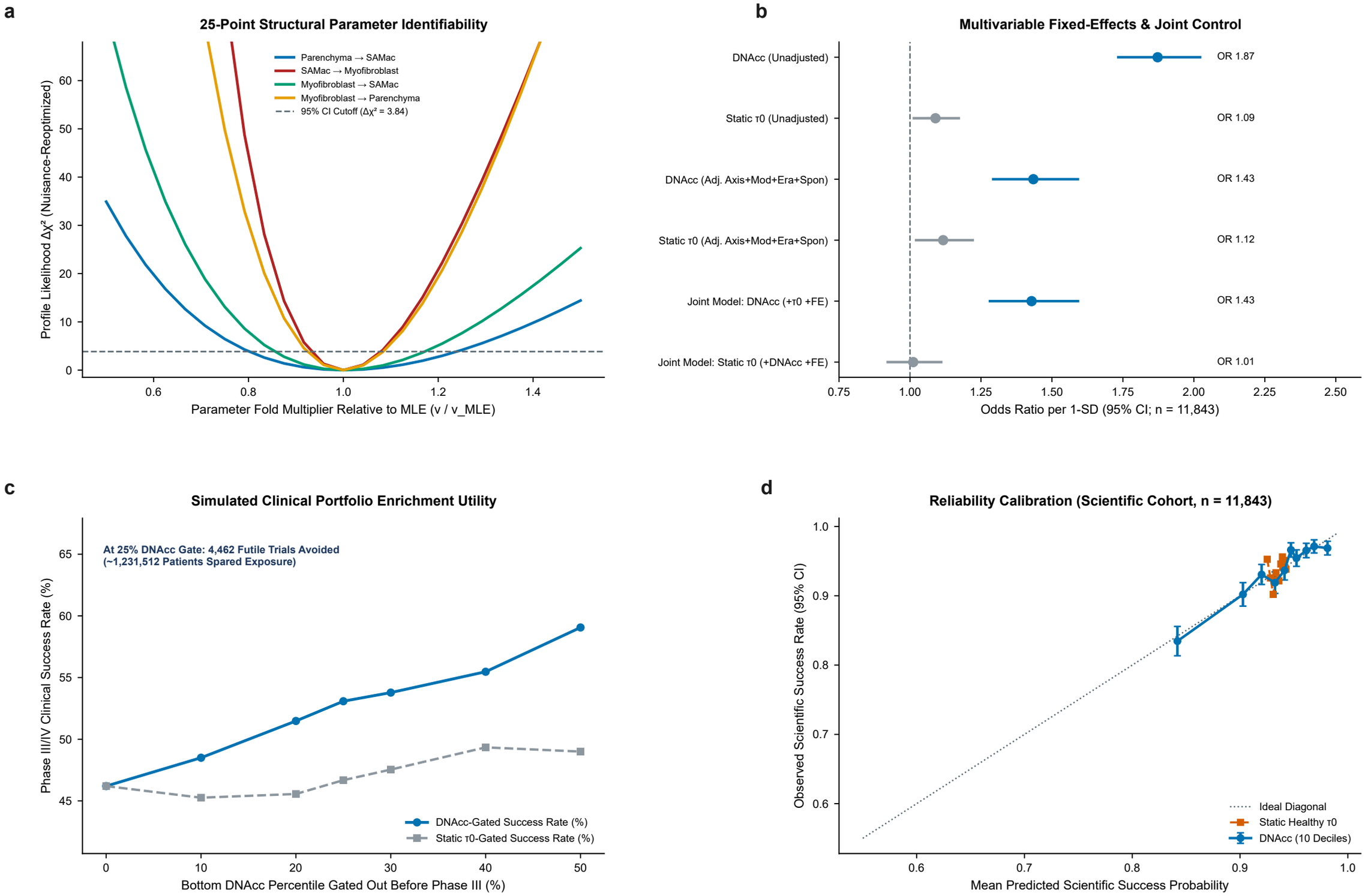

a
25-Point Structural Parameter Identifiability
Parenchyma → SAMac
SAMac → Myofibroblast
Myofibroblast → SAMac
Myofibroblast → Parenchyma
95% CI Cutoff (Δχ² = 3.84)
Profile Likelihood Δχ² (Nuisance-Reoptimized)
Parameter Fold Multiplier Relative to MLE (v / v_MLE)
b
Multivariable Fixed-Effects & Joint Control
DNAcc (Unadjusted)
OR 1.87
Static τ0 (Unadjusted)
OR 1.09
DNAcc (Adj. Axis+Mod+Era+Spon)
OR 1.43
Static τ0 (Adj. Axis+Mod+Era+Spon)
OR 1.12
Joint Model: DNAcc (+τ0 +FE)
OR 1.43
Joint Model: Static τ0 (+DNAcc +FE)
OR 1.01
Odds Ratio per 1-SD (95% CI; n = 11,843)
c
Simulated Clinical Portfolio Enrichment Utility
At 25% DNAcc Gate: 4,462 Futile Trials Avoided
(~1,231,512 Patients Spared Exposure)
DNAcc-Gated Success Rate (%)
Static τ0-Gated Success Rate (%)
Phase III/IV Clinical Success Rate (%)
Bottom DNAcc Percentile Gated Out Before Phase III (%)
d
Reliability Calibration (Scientific Cohort, n = 11,843)
Ideal Diagonal
Static Healthy τ0
DNAcc (10 Deciles)
Observed Scientific Success Rate (95% CI)
Mean Predicted Scientific Success Probability